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Negative Binomial Kumaraswamy-G Cure Rate Regression Model

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Abstract

In survival analysis, the presence of elements not susceptible to the event of interest is very common. These elements lead to what is called a fraction cure, cure rate, or even long-term survivors. In this paper, we propose a unified approach using the negative binomial distribution for modeling cure rates under the Kumaraswamy family of distributions. The estimation is made by maximum likelihood. We checked the maximum likelihood asymptotic properties through some simulation setups. Furthermore, we propose an estimation strategy based on the Negative Binomial Kumaraswamy-G generalized linear model. Finally, we illustrate the distributions proposed using a real data set related to health risk.

Keywords: long-term survivors; Kumaraswamy family; survival analysis; negative binomial distribution; generalized linear model